

Roshan Kumar Gupta

Research Engineer & Systems Architect

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🌐 tars.earth

Education

2025–2029 **Bachelor of Technology in Computer Science**, Chandigarh University, Mohali, Expected
Relevant Coursework: High-Performance Computing, Low-Level Systems, Parallel Architectures, Database Systems.

Experience

- Jul 2026–Present **Founder & Lead Architect**, *Earthos Platform*, earthos.shop
- Deployed the production storefront and interactive reader workspace (earthos.shop) for *The AGI Question* manuscript using Next.js 16 (React 19) and Supabase PostgreSQL.
 - Engineered automated multi-warehouse nearest-node shipping routing algorithms and integrated Shiprocket APIs to dynamically optimize logistics.
 - Developed a secure PDF generation pipeline with real-time DRM watermark fingerprinting and cryptographic user-state injection.
- Dec 2025–Jan 2026 **L3 IT Intern**, *SMG Electric Scooters Ltd*, Chandigarh
- Designed and developed a custom internal CRM portal, streamlining logistics and vendor development workflows for enterprise-scale operations.
 - Collaborated with cross-functional teams to identify infrastructure bottlenecks, reducing manual processing time through automation.
- Oct 2025–Present **Student Representative**, *Chandigarh University*, Mohali
- Act as primary student-admin liaison, managing high-stakes communication to resolve complex infrastructure and academic issues.
 - Facilitated multiple departmental town-halls, ensuring transparent information flow and collaborative problem-solving across 400+ student cohorts.

System Engineering

- Project Earthos **Project Earthos (Synapse Arch)** | *Autonomous Cognitive Substrate*
- Architected and implemented Synapse Arch, a neural-symbolic cognitive substrate combining event-sourced memory hydration and global workspace attention (GWT) coalitions.
 - Implemented AST and import locks to block non-deterministic floating-point namespaces, enforcing strict byte-level execution parity across distinct CPU platforms.
- FieldChain **FieldChain** | github.com/rk-roshan-kr/FieldChain
- Secured a \$1,500 research grant; engineered a GPU-centric storage integrity layer sustaining 7GB/s sequential writes via PCIe 4.0, outperforming specialized DPUs by 4.2x (166.86 GiB/s peak bandwidth) via Vulkan.
 - Selected to present research paper at the IEEE CCNCPS 2026 conference in Dubai, UAE.
- Project TARS **Project TARS Ecosystem** | tars.earth
- Secured \$5,500 Emergent Ventures grant; designed the TARS Core physics-constrained validation framework for sparse two-transit exoplanet candidate vetting, achieving 70.1% precision on NASA TESS data.
 - Formulated architectural specifications for the TARS-EX (physics-aware ensemble ML) research branch.
 - Engineered *tars_infra*, a parallel FITS data ingestion engine utilizing multi-drive spanning to download and index 150,000+ files from the MAST database.
- QUANTA **Thesis QUANTA** | Working Paper
- Defined an operational info-horizon for NISQ circuits via entropic noise metrics; formulated exponential scaling law and mapped the "Undead Zone" to balance state extractability against environmental decoherence.
- HSA* / Arch **HSA* Algorithm & Roshan Arch**
- Formulated routing engine using inverse square law safety weights, achieving pathfinding in 97μs (140x faster than Dijkstra).
 - Architected a custom deterministic memory framework in C++ to bypass garbage collection pauses and ensure predictable real-time runtimes.
- Cryptic **Cryptic** | ZeroToOne Hackathon Winner
- Served as Frontend Lead for a DeFi Treasury dashboard, managing UI/UX architecture and 3D WebGL spatial liquidity flow visualizations.

Technical Skills

Low-Level Systems CUDA, Vulkan, PTX, C++, Deterministic Runtimes, Memory Architectures.
React (Architecture), PyTorch (Physics-ML), Figma (UI Primitives).

Honors & Certifications

Awards Winner, ZeroToOne Hackathon; Winner, Zinnovatio Hackathon; 15+ Hackathons participation.
Certs Scientific Writing (Polytechnique), Azure AI Cert (Vision, NLP).

Extracurricular Activities

Hackathons Highly active in the global hackathon circuit (15+ events), focus on building collaborative open-source tools.

Interests Exploring the intersection of quantum noise limits, signal processing, and low-level GPU acceleration.